Filip Ramazan

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EDUCATION

The Davidson Academy of Nevada, *4.0/4.0 UW GPA* *2020 – Present*

- Relevant coursework: Calculus I/II, Advanced Biology, Advanced Physics, Machine Learning, Advanced Chemistry, Programming in Python, Programming with Java, Quantum Mechanics, Astronomy.

University of Nevada, Reno, *4.0/4.0 UW GPA* (concurrent enrollment with high school) *2020 – Present*

- Selected undergraduate coursework: Differential Equations (MATH 285), Multivariable Calculus (MATH 283), Linear Algebra (MATH 330), Introduction to Proofs (MATH 295), Probability (STAT 461), General Chemistry (CHEM 121L & 122L), Principles of Biological Investigation Lab and Research Writing (BIOL 192), Molecular Genetics (BIOL 304), Neurobiology (BIOL 475*), Peer Leadership in Biology (BIOL 495*) [*indicates dual undergraduate/graduate-level classes].
- Selected graduate coursework: Molecular Biology (BCH 705, gene-editing, genomics, and gene expression), Machine Learning (STAT 760, neural networks, regression, transformers).

RESEARCH AND WORK EXPERIENCE

Yim Genomics Laboratory, *Researcher* – UNR *August 2024 – Present*

- Collaborated with Dr. Won Yim and Dr. John Lomas to develop and train a transformer model to predict gene structure and alternative splicing directly from a DNA sequence.
- Develop a DNA tandem-repeat finder algorithm and train models to find transposons.

Nigam Shah Lab, *Researcher* – Stanford, CA *February 2025 – Present*

- Built and integrated, in collaboration with Akshay Swaminathan, LLM tools for the Vandrevala Foundation, helping provide faster and better free mental health counseling to hundreds of patients daily in low-resource locations in India.
- Train small custom LLMs to run on-device and offline to improve quality of digital SOAP notes written by tens of eHealth Africa's community health workers and pharmacists in Nigeria.

Peer Instruction Program, *Leader* – UNR *November 2024 – Present*

- Coordinate with teaching assistant team to design curriculum for BIOL 304 (Molecular Genetics).
- Teach a weekly discussion session of 20 students with difficult upper division level genetics concepts, from DNA sequencing, CRISPR, RNAi to gene linkage and mapping.
- Reviewed and edited textbook chapters for *Genetic Analysis: An Integrated Approach* 4e alongside Dr. Pamela Sandstrom.

Math Center, *Senior Tutor* – UNR *July 2023 – April 2024*

- Assisted walk-in students in developing mathematical and statistical skills, on topics ranging from Pre-Algebra to Differential Equations.
- Led multiple exam review sessions for UNR and community college students.

CAMPUS & COMMUNITY ENGAGEMENT

Code 4 Romania, *Volunteer* *May – September 2025*

- Worked with developer team to design and launch apps to help nonprofits and foreign governments with vote monitoring systems and refugee care.

Changing Dynamics, *Field-volunteer* *July – October 2024*

- Canvassed nearly 700 homes in Reno, NV and collected dozens of petition signatures during the 2024 election cycle.

sciencebowl.org, *Developer and Maintainer* *August 2025 – Present*

- Created the largest ever database of Science Bowl questions in history and developed a free website to use this database with multiplayer buzzer features, practice modes, round generators, and a team management platform.
- Marketed website to reach 1,100+ of weekly users, helping underserved teams to practice without buying expensive buzzer systems and prep materials.

Davidson Academy GECKO, *Club Vice President* *January 2025 – Present*

- Fundraised $1,000 per semester for *Keep Tahoe Blue* and *Keep Truckee Meadows Beautiful*.
- Spearheaded DA e-waste drive, recycling hundreds of pounds of li-ion batteries and old electronics with Redwood Materials.
- Collaborated with UNR's Facilities Services to launch a campus-wide e-waste recycling program.
- Working to establish school-wide food-waste composting program.

COMPETITIONS & AWARDS

Two-time National Biology Bowl Champion *2021 – 2023*
- Captained the Davidson Academy's team to 1st place at nationals, organized weekly practices and wrote study materials.
- Recruited 8 club members as club president and coached a second team to win 4th place nationally.

USA Brain Bee, Rutgers University, NJ *2025*
- Placed in top 15 nationally at Brain Bee nationals, after winning the Nevada Brain Bee.
- Helping coordinate the 2026 National and Nevada Brain Bee alongside Dr. Amy Altick of UNR.

US Department of Energy National Science Bowl, Las Vegas, NV & Washington, D.C. *2023 – Present*
- Placed 5th nationally in Washington D.C. with a $1,000 prize in 2023 (MS) and 8th nationally with a $1,000 prize in 2025 (HS).
- Won Nevada regional competitions in 2023, 2024, and 2025, each with a $5,000 award.

UToronto National Biology Competition, Online *2024, 2025*
- Placed in top 40 in 2024 and 2025 out of 3,000+ competitors internationally.

MIT Science Bowl Invitational, Cambridge, MA *2024, 2025*
- Placed in top 8 out of the best 48 teams nationally.

Stanford Science Bowl Invitational, Palo Alto, CA *2025*
- Placed in top 8 out of the best 48 teams nationally.

Berkeley Science Bowl Invitational, Berkeley, CA *2025*
- Placed 1st out of 48 of the best teams nationally.

2-time USA Biology Olympiad Semifinalist and Certificate of Merit *2025*
- Placed in first 225 scores on the United State Biology Olympiad Semifinal exam.

2025 Congressional App Challenge 2nd Place *2025*

2026 Carson Scholar *2026*
- Received $1,250 college scholarship and one of only 4 scholars to be given the Michael S. Hollins Humanitarian Award for exceptional public service.

LANGUAGES, SKILLS, AND TEST SCORES

Programming

Python (emphasis in data science), Java (and Android development), LaTeX, HTML, JS, CSS, React, C++/C

Selected Test Scores

Biology/Calculus AB & BC/Spanish Language & Culture/Chemistry: 5, PSAT: 1500, SAT: 1550, ACT: 35

Languages

English and Romanian (native), Spanish and French (full professional proficiency)

PUBLICATIONS

Swaminathan, A.*, Ramazan, F.*, Lopez, I.*, Wang, K., Shah, N. *A multi-turn LLM evaluation framework for copilot LLMs*, 2025, *Lancet Global Health*. (submitted for publication)

Lomas, J. S., Ramazan, F., Cushman, J. C., Tang, H., & Yim, W. C. *TransGenic: A transformer-based framework for direct DNA-to-annotation translation*, 2025, *Plant Communications.* (accepted for publication)

* indicates co-first author